10. Kanban
Reduce waste

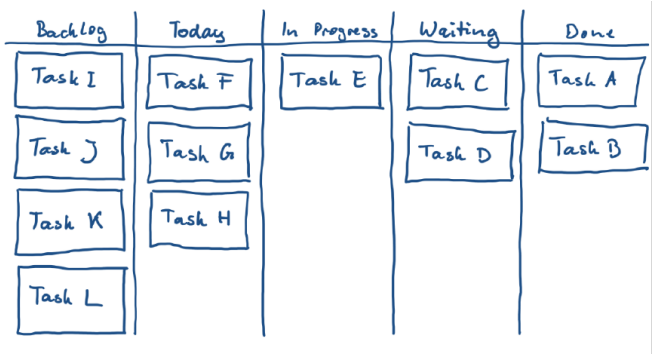
"Software Development Methodologies"
Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Kanban?



Kanban?

- From Japanese, meaning "visual signal“
- Developed by Toyota, in the late 40’s by Taiichi Ohno



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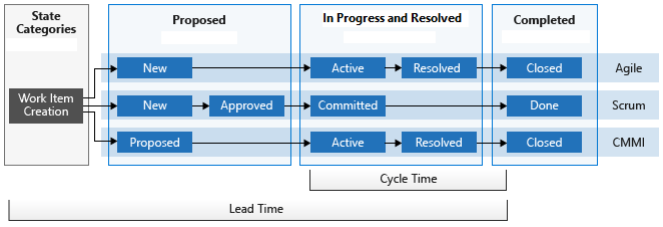
Basic principles

- Visualize the workflow
  - Split the work into pieces, write each item on a card and put on the wall
  - Use named columns to illustrate where each item is in the workflow
- Limit the work in progress (WIP)
  - assign explicit limits to how many items may be in progress at each workflow state.
- Measure the **lead time** (average time to complete one item from request to delivery)
- Measure the **cycle time** (time from work start to ready for delivery)
  - optimize the process to make lead time/cycle time as small and predictable as possible.

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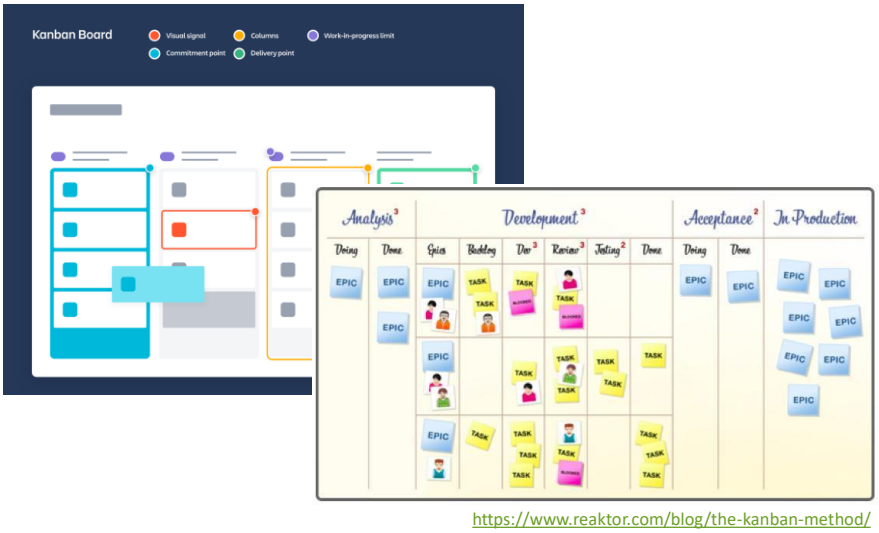
Basic principles



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Kanban Board examples

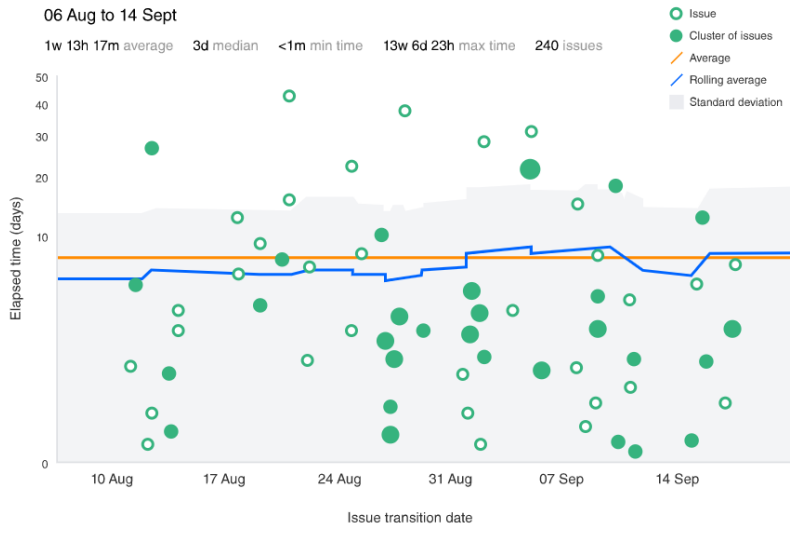


https://www.reaktor.com/blog/the-kanban-method/

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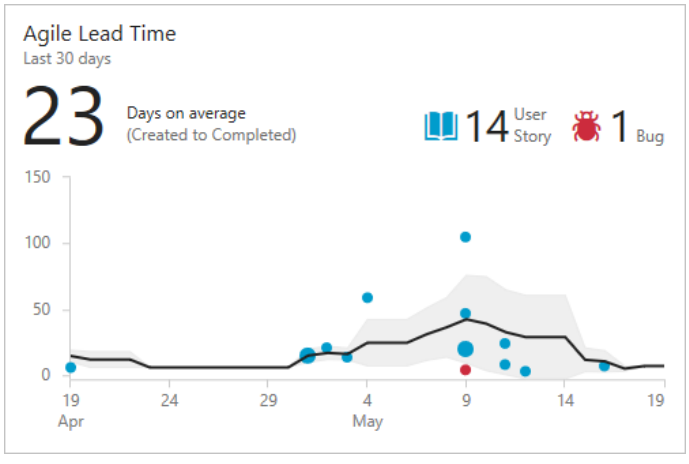
Visual metrics example - cycle time for each issue



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Visual metrics example – lead time for each issue

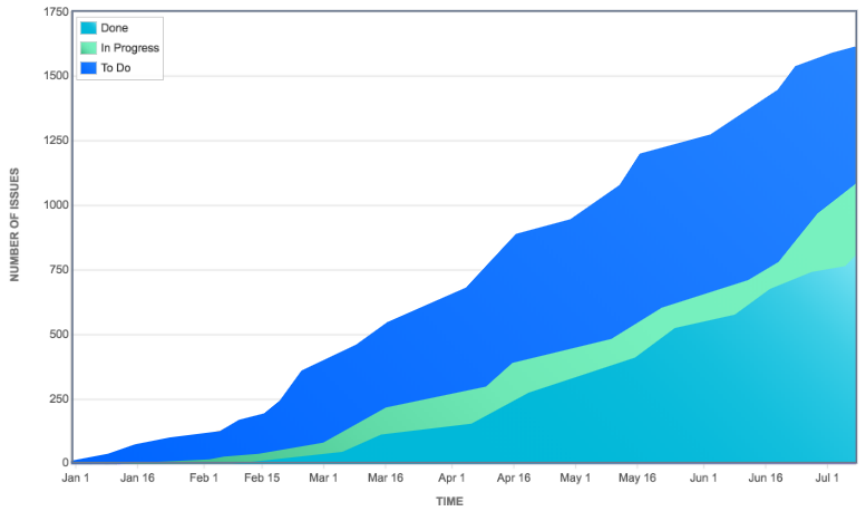


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Visual metrics example - cumulative flow diagram

(number of issues in each state)



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Benefits of Kanban

- Planning flexibility
  - A Kanban team is only focused on the work that's actively in progress.
  - No need for fixed-length iterations.
- Shortened time cycles
  - Cycle time is a key metric. By optimizing cycle time, the team can confidently forecast the delivery of future work.
- Fewer bottlenecks
  - Limiting the amount of WIP reduces context switching, and improves efficiency
- Visual metrics
  - Charts provide a visual mechanism for teams to ensure they're continuing to improve.

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SCRUMBAN

- Scrum + Kanban
  - Use the prescriptive nature of Scrum to be Agile.
  - Use the process improvement of Kanban to allow the team to continually improve its process.
  - Average lead time and cycle time will become the primary focus of performance
  - Rather than estimating a scope of work for every iteration, just make the backlog a fixed size
  - Iteration planning at regular intervals, synchronized with review and retrospective
  - The goal of planning is to fill the slots available – not fill all of the slots, and not determine the number of slots

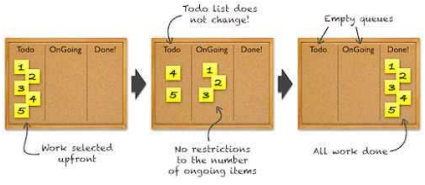
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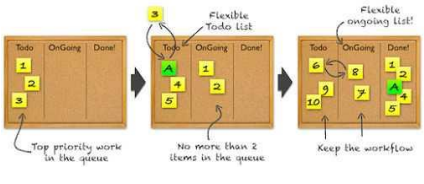
Scrumban advantages over Scrum

- No burn-down
- No fixed iterations
- No need to estimate (similar size work)
- Team can be specialized
- Changes are added as needed, don't have to wait for the next Sprint

Scrum

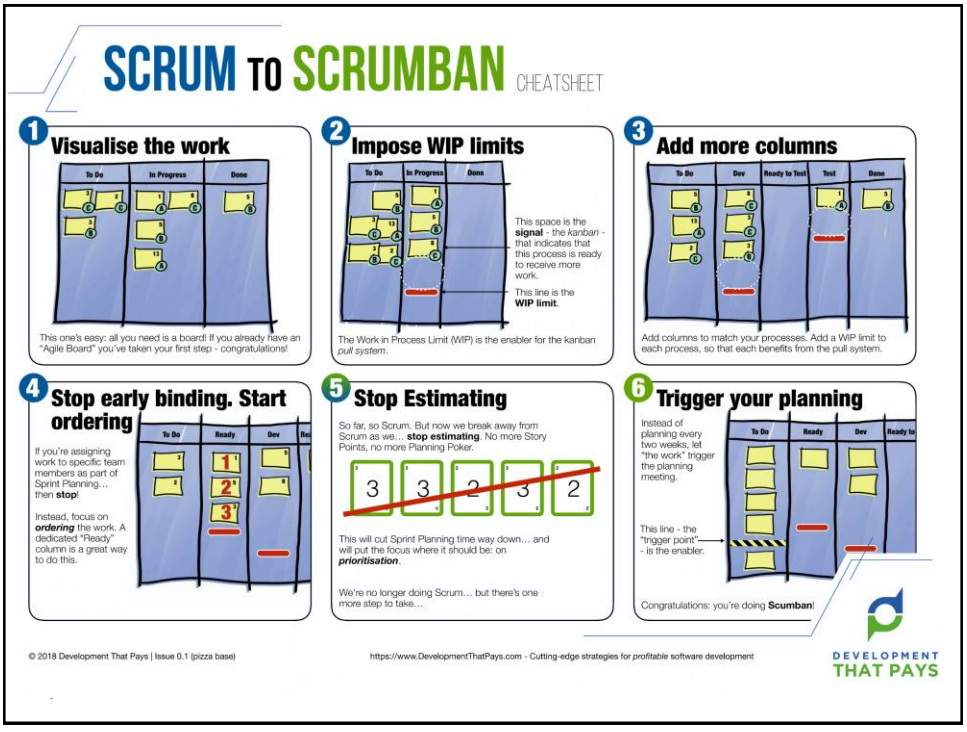


Kanban



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